

Apache OpenOffice

Version 4.1

Draw Guide

AOO Documentation Team

Chapter **5**

Combining Multiple Objects

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Note for Mac Users

Some keystrokes and menu items differ on a Mac from those used in Windows and Linux. The table below gives some common substitutions for the instructions in this chapter. For a more detailed list, see the application Help.

<i>Windows/Linux</i>	<i>Mac equivalent</i>	<i>Effect</i>
Tools > Options menu selection	OpenOffice > Preferences	Access setup options
<i>Right-click</i>	<i>Control+click</i>	Open context menu
<i>Ctrl</i> (Control)	⌘ (Command)	Used with other keys
<i>F5</i>	<i>Shift+⌘+F5</i>	Open the Navigator
<i>F11</i>	<i>⌘+T</i>	Open Styles & Formatting window

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Introduction: Grouping and Combining Objects

With Draw, you can combine drawing objects together in two distinct ways: *grouping* and *combining*. These two methods allow you to treat multiple objects as a single unit or to merge objects to form a new shape.

Grouping is similar to placing objects in a container. You can move them as a group and apply global changes to them. A group can always be undone, and its constituent objects can always be manipulated separately.

A *combination* is a permanent fusion of objects, resulting in a new object. The original objects are no longer available as individual entities, and the operation is not reversible.

Tip

To select multiple objects, click on each object while holding down the *Shift* key or click the **Select** icon (on the Drawing toolbar) and draw a rectangle around the objects.

Grouping Objects

Grouping by Common Selection

When several objects are selected, any operations you carry out are applied to all of the objects. For example, you can rotate a group of objects in its entirety. Groups obtained through the common selection of several objects are undone as soon as you click outside the group. However, you can choose to maintain a grouping so that the selected objects will remain together after they are deselected.

Maintaining Groups and Undoing Groups

To group objects, first select the objects, then right-click and choose **Group** from the pop-up menu, as shown in Figure 1. You can also use the keyboard shortcut *Control+Shift+G* or choose **Modify > Group** from the menu bar.

When objects are grouped, any editing operations performed on that group are applied to all its members. If you click on one member of the group, the whole group is selected.

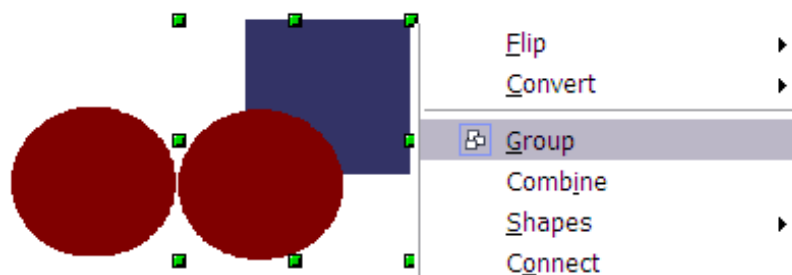


Figure 1: Grouping objects using the right-click menu

The objects of a group retain their unique individual properties. To undo a group, right-click and choose **Ungroup** from the pop-up menu, use the keyboard shortcut *Control+Alt+Shift+G*, or choose **Modify > Ungroup** from the menu bar.

Editing Individual Objects in a Group

You can edit a member of a group individually without breaking up the group. To do this, right-click and choose **Enter group** or double-click on the group (Figure 2).

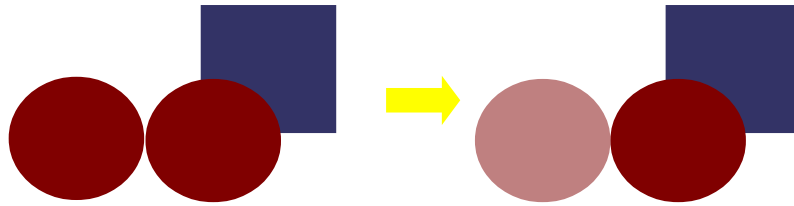


Figure 2: When you enter a group, objects outside the group cannot be selected and appear pale

Once inside the group, click on any object to edit it (Figure 3).

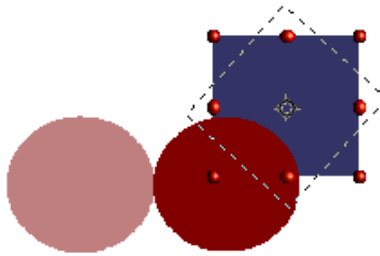


Figure 3: Editing an object inside a group

To leave this mode, right-click and choose **Exit group**, or double-click outside the group.

Nesting Groups

You can create “nested” groups, that is, groups within groups. In this case, Draw keeps the initial group hierarchy. If you ungroup a group made of other groups, you are left with individual groups that you can then decompose further, until only single objects remain.

Combining Objects

In contrast to grouping functions, *combinations* create a new object; it is not possible to edit an individual constituent afterwards. Select a collection of objects, then right-click and choose **Combine** from the pop-up menu or use the keyboard shortcut *Control+Shift+K*. The result of this operation is shown in Figure 4.

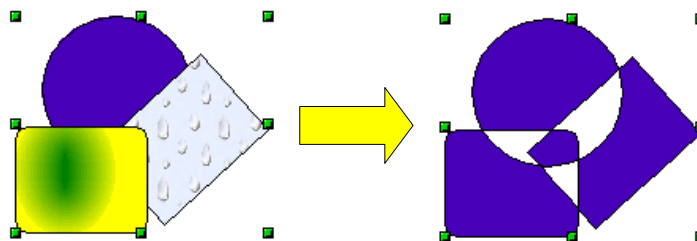


Figure 4: Combining objects

At first glance, the appearance of the combined objects can seem rather surprising. Understanding the rules governing combinations in Draw will clarify things.

- The attributes (for example, area fill) of the resulting object are those of the object furthest back. In this example, it is the circle.
- Where the objects overlap, the overlapping zone is either filled or empty, depending on the number of overlaps. When the number of overlaps is even, you get an empty space; when the number is odd, you get a filled area, see Figure 5.

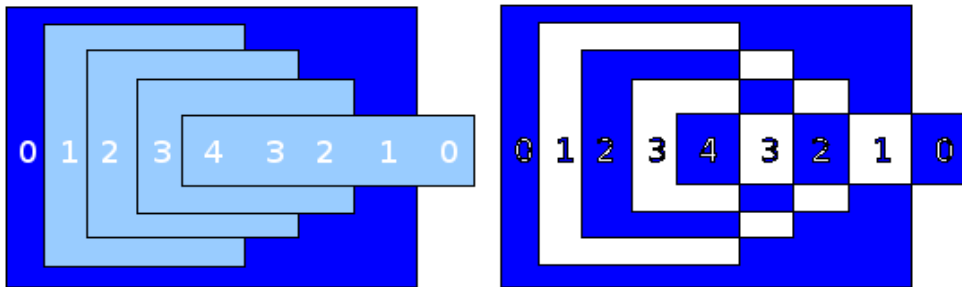


Figure 5: Odd numbered overlaps are filled, even numbered overlaps are empty

Tip

You can reorder objects before combining them so that they are positioned further back or further forward. Right-click on the object and select **Arrangement** from the pop-up menu. See also “Aids for Positioning Objects” on page 8.

An object that is a combination can be broken up again by selecting **Modify > Split** or using the keyboard shortcut *Control+Alt+Shift+K*. The combination can also be broken up into its constituent parts by selecting **Modify > Break**. These are not an exact equivalent to ungrouping, as the objects were converted to Bézier curves when they were combined. An object that is split will, for example, have an ellipse converted to a Bézier curve.

An object that is broken will consist of a set of base-level objects (for example, a rectangle will be broken into four separate lines), and the original formatting—line thickness, colors, and area properties—is not retained.

Connecting Lines

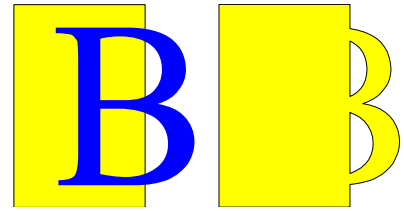
Lines may be connected by selecting them and using **Modify > Connect**. The connections established are based on joining the ends of pairs of lines with minimum separation. Once a group of lines is connected, you can “close” the remaining gap by selecting the line, right-clicking, and choosing **Close Object**. The resultant object is filled with the default fill. If you modified the line properties (color, weight), these are retained in the resulting polygon.

Merging, Subtracting, or Intersecting Shapes

After you have selected more than one object, the Merge, Subtract, and Intersect functions can be reached in the **Modify > Shapes** menu or through the group’s right-click menu under the heading **Shapes**.

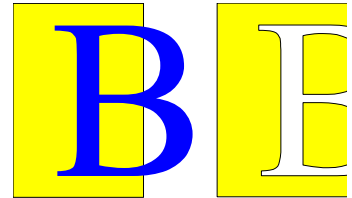
Merge

When you merge objects, the new object covers the entire surface of the original objects, representing the *union* of the objects. As was the case with combining, the fill of the merged object is determined by the fill of the object furthest in the background. Other properties are not transferred to the new object.



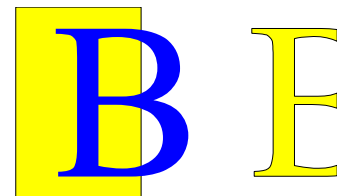
Subtract

When you subtract, the object in front is subtracted from the object behind it.



Intersect

When you intersect two objects, you get only the area covered by both objects.



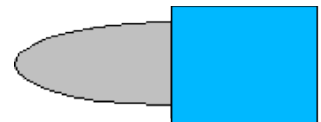
Tip

For more information on moving objects to the front or back, see page 8.

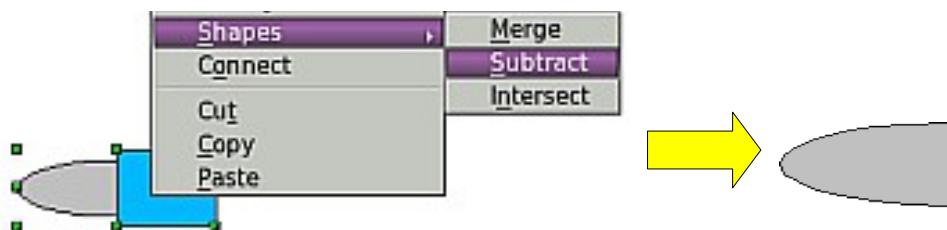
Practical Example

The following example shows how to use the merge functions to create a complex shape—a knife with a wooden handle.

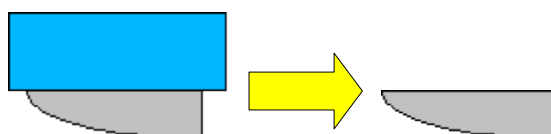
- 1) Draw an ellipse and then a rectangle overlapping half of its width.



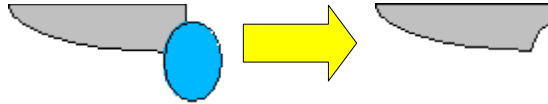
- 2) Select both shapes, right-click, and choose **Shapes > Subtract** from the pop-up menu. The result should resemble the shape to the right, as shown below:



- 3) Draw another rectangle and put it over the top half of the ellipse. Then subtract again.



- 4) Draw a small ellipse covering just the lower right corner and subtract again.



- 5) The knife blade is now complete. To make the handle, draw a rectangle and an ellipse as shown.



- 6) *Merge* the shapes.



- 7) *Group* this image with the previous one.



Aids for Positioning Objects

Draw has various tools to help you arrange objects with respect to each other. Here we explore some of the more important ones.

Moving an Object to the Front or to the Back

When you combine or merge objects, the result varies drastically depending on which object is “in front” and which one is “behind”. Figure 6 illustrates the difference.

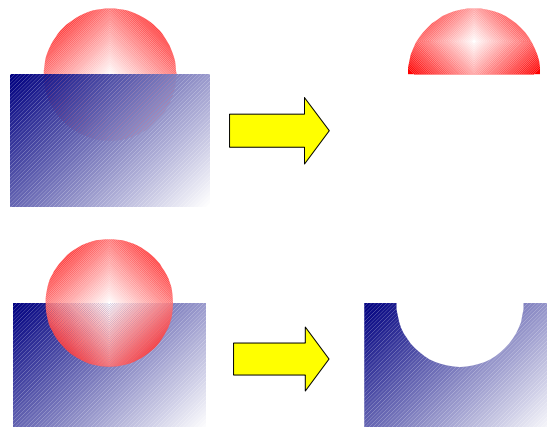


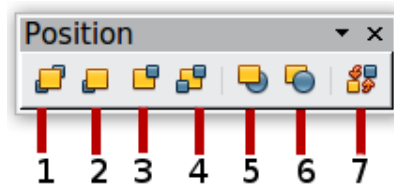
Figure 6: Subtracting objects: the result varies depending on which object is in front.

If you imagine several objects stacked one on top of another, the one referred to as “Front” is the one on top of the stack. The one referred to as “Back” is the one at the very bottom of the stack.

First, select an object. Then, on the Drawing toolbar, long-click the Arrange icon



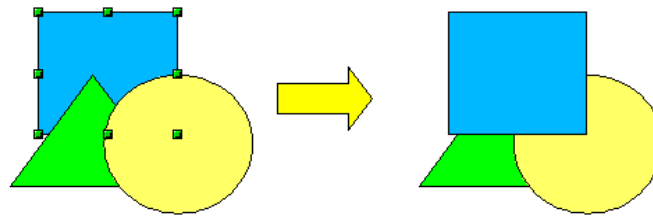
to open the Position toolbar (Figure 7). This toolbar can be torn off and made to float.



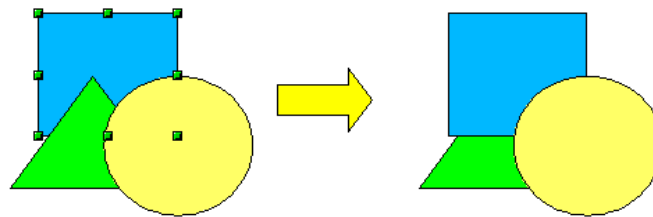
- | | |
|-------------------------|-----------------------------|
| 1 Bring to Front | 5 In Front of Object |
| 2 Bring Forward | 6 Behind Object |
| 3 Send Backward | 7 Reverse |
| 4 Send to Back | |

Figure 7: The floating Position toolbar

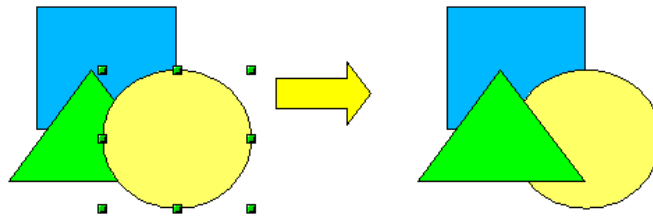
 Brings the selected object to the front of the group.



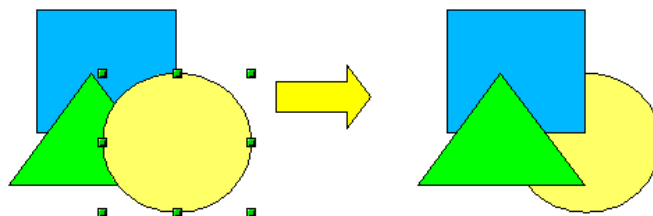
 Brings the selected object one step forward.



 Sends the selected object one step backwards.



 Sends the selected object to the back of the group.

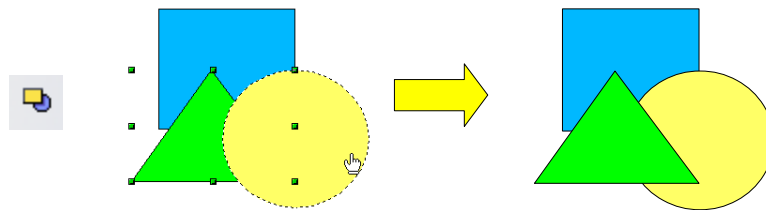


 Moves the selected object in front of another chosen object.

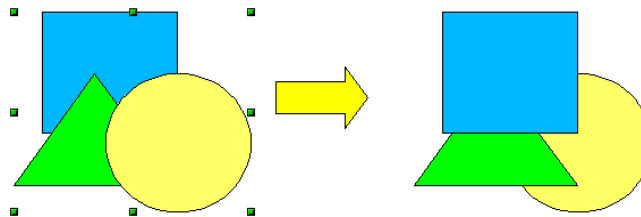
 Moves the selected object behind another chosen object.

First, click on the object you want to change in position. Next, click on the icon and then move the mouse over the other object. The mouse cursor changes into a hand,

and the outline of the second object becomes a dashed line (see below). Clicking on this object completes the rearrangement of position.

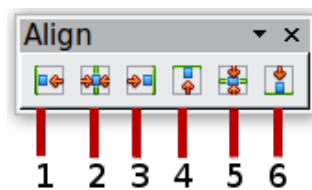


Reverses the order of the selected objects (this option is grayed out if only one object is selected).



Aligning Objects

Select two objects and long-click the Alignment icon  on the Drawing toolbar to open the **Align** toolbar (Figure 8). This toolbar can be torn off and made to float. The tools help you position objects with respect to each other.

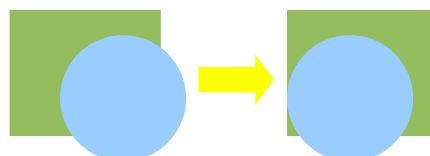


1 Align left	3 Align right	5 Center vertically
2 Center horizontally	4 Align top	6 Align bottom

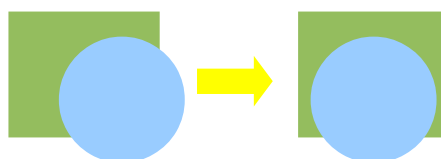
Figure 8: The floating Align toolbar



Align left.

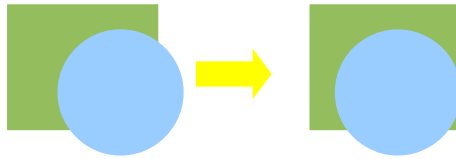


Center horizontally.

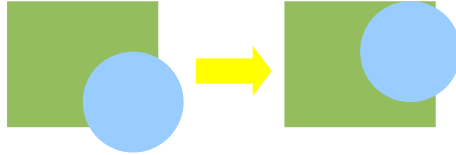




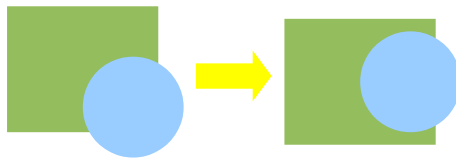
Align right.



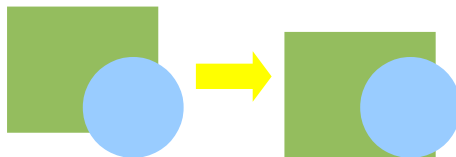
Align top.



Center vertically.



Align bottom.



If you have selected only one object, clicking the various buttons on the Align toolbar positions the object in relation to the drawing page.

Distributing the Distance and Space Between Objects

Select at least three objects. Then you'll be able to select the **Distribution** option from the **Modify** menu or from the right-click menu. The Distribution dialog (Figure 9) contains several options, which are described below.

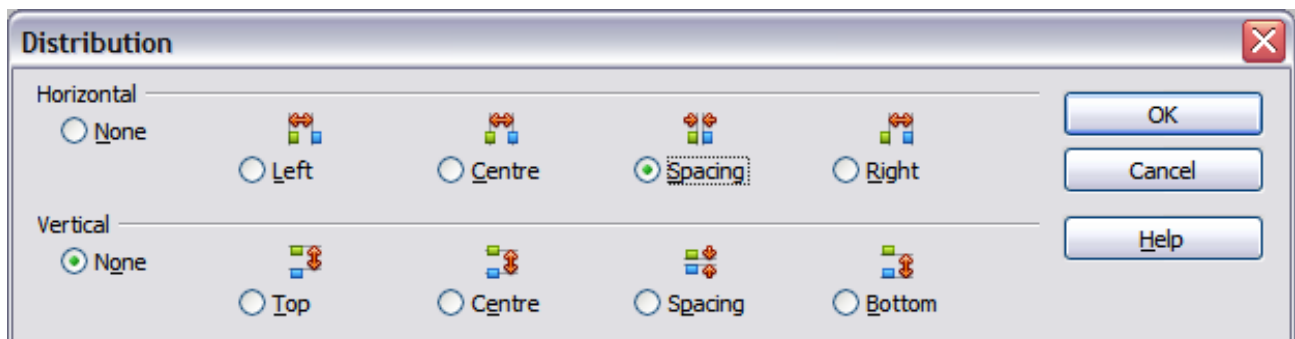


Figure 9: Distribution of spacing between objects

Horizontal distribution

None. Does not distribute the objects horizontally.

Left. The left edges of the objects are evenly spaced from one another.

Center. The horizontal centers of the objects are evenly spaced from one another.

Spacing. The horizontal spacing between the objects is evenly distributed.

Right. The right edges of the objects are evenly spaced from one another.

Vertical distribution

None. Does not distribute the objects vertically.

Top. The top edges of the objects are evenly spaced from one another.

Center. The vertical centers of the objects are evenly spaced from one another.

Spacing. The vertical spacing between the objects is evenly distributed.

Bottom. The bottom edges of the objects are evenly spaced from one another.

Figure 10 shows four objects before horizontal spacing distribution.



Figure 10: Before distribution

The result of the selections made in Figure 9 is shown in Figure 11.

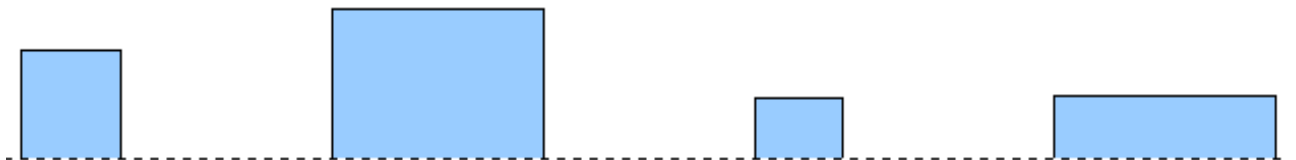


Figure 11: After distribution: change to equal horizontal spacing, no vertical change